Batu Ozturkler

650-680-6618 | batumozturkler@gmail.com | batuozt.github.io | Google Scholar

Research Profile

Research scientist specializing in **LLM post-training, learned agent orchestration, and scalable evaluation**, with experience translating research into production. Recent work spans RL environments for training long-horizon coding orchestrators, synthetic-data generation for post-training, and calibrated confidence estimation. Stanford Ph.D. with research in probabilistic LLM reasoning, diffusion models, and computer vision, emphasizing efficiency and robustness. Interested in applying this foundation to **AI systems that accelerate and automate scientific research**.

Experience

Senior Research Scientist Aug 2026 – Present
Research Scientist 2 Mar 2026 – Jul 2026
Microsoft CodeAI *Mountain View, CA*

- **RL Training for Agent Orchestration:** Built an RL environment integrating a coding-agent harness and user simulator to generate multi-turn rollouts and rewards. Trained small coordinator LLMs on these rollouts to compose tool-using workflows and route steps across models for GitHub Copilot Fusion. Evaluated at its max setting in Copilot CLI, the released system performed within approximately two index points of Claude Code with Opus 5.5 (max) at **51% lower API cost per task** on the Artificial Analysis Coding Agent Index v2.0 (Oct. 2026).
- **Production Model Training and Deployment:** Led development of a fine-tuned ModernBERT credential scanner (blog post) deployed across GitHub's internal and external products at the scale of all GitHub push traffic. Achieved higher precision than multi-LLM workflows by **over 11.5%** in online A/B tests, while reducing inference cost by **over 98%** and latency by **over 86%**.

Applied Scientist 2 May 2024 – Feb 2026
Microsoft *Mountain View, CA*

- **Multilingual LLM Training and Synthetic Data:** Co-developed multilingual training for GPT-4o spanning mid-training and post-training (SFT, DPO, RLHF). Built large-scale benchmarks, scalable evaluation pipelines, and synthetic-data generation systems that improved multilingual model performance.
- **Model Uncertainty and RL-Based Post-Training:** Developed post-training methods for calibrated confidence estimation using auxiliary predictors and RL-based optimization with calibrated rewards, improving hallucination detection and reliability by 10% AUROC in production systems including vision-language models for structured extraction from long documents.

Postdoctoral Scholar Mar 2024 – May 2024
Research Assistant Sep 2019 – Jan 2024
Stanford University *Stanford, CA*

- Developed memory-efficient training methods for high-dimensional MRI reconstruction and self-supervised-learning-based augmentation to improve robustness under signal-to-noise and motion distribution shifts.
- Developed equivalent convex formulations of non-convex neural networks to provide robustness, interpretability, and convergence guarantees.

Research Intern Sep 2022 – Apr 2023
NVIDIA *Santa Clara, CA*

- Improved the robustness and efficiency of diffusion models for MRI reconstruction under distribution shifts using SURE and variational inference.

Research Intern Jun 2022 – Aug 2022
Microsoft Research *Redmond, WA*

- **LLM Reasoning and Failure Analysis:** Developed a probabilistic inference method that improved LLM reasoning and robustness to prompt design; characterized failure modes of in-context learning.

Education

Stanford University Sep 2019 – Jan 2024
Ph.D. in Electrical Engineering; GPA: 3.99/4.00 *Stanford, CA*
- Advisors: Prof. John Pauly and Prof. Mert Pilanci
- Thesis: Efficient and robust deep learning for medical imaging and natural language processing

Stanford University Sep 2019 – Jun 2021
M.S. in Electrical Engineering; GPA: 3.99/4.00 *Stanford, CA*

Middle East Technical University (METU) Aug 2015 – Jun 2019
B.S. in Electrical Engineering; GPA: 3.96/4.00, Rank: ***3/413*** *Ankara, Turkey*

Selected Publications

1. **Ozturkler, B.M.**, Malkin, N., Wang, Z., and Jojic, N. ThinkSum: Probabilistic reasoning over sets using large language models. **ACL 2023**. arXiv:2210.01293.
2. Sahiner, A., Ergen, T., **Ozturkler, B.M.**, Pauly, J., Mardani, M., and Pilanci, M. Scaling Convex Neural Networks with Burer-Monteiro Factorization. **ICLR 2024**.
3. Sahiner, A., Ergen, T., **Ozturkler, B.M.**, Pauly, J., Mardani, M., and Pilanci, M. Unraveling Attention via Convex Duality: Analysis and Interpretations of Vision Transformers. **ICML 2022**.
4. **Ozturkler, B.M.**, Mardani, M., Vahdat, A., Kautz, J., and Pauly, J. RED-diff: Regularization by Denoising Diffusion Process for MRI Reconstruction. **NeurIPS 2023 Workshop on Deep Learning and Inverse Problems**.
5. Desai, A.D.*, Gunel, B.*, **Ozturkler, B.M.**, et al. VORTEX: Physics-Driven Data Augmentations for Consistency Training for Robust Accelerated MRI Reconstruction. **MIDL 2022 Best Paper Award**.
6. **Ozturkler, B.M.**, Liu, C., Eckart, B., Mardani, M., Song, J., and Kautz, J. SMRD: SURE-based Robust MRI Reconstruction with Diffusion Models. **MICCAI 2023** (early accept, top 15% of all submissions).
7. Sahiner, A.*, Ergen, T.*, **Ozturkler, B.M.**, Bartan, B., Pauly, J., Mardani, M., and Pilanci, M. Hidden Convexity of Wasserstein GANs: Interpretable Generative Models with Closed-Form Solutions. **ICLR 2022** (* equal contribution).
8. Ergen, T.*, Sahiner, A.*, **Ozturkler, B.M.**, Pauly, J., Mardani, M., and Pilanci, M. Demystifying Batch Normalization in ReLU Networks: Equivalent Convex Optimization Models and Implicit Regularization. **ICLR 2022** (* equal contribution).
9. Sahiner, A., Mardani, M., **Ozturkler, B.M.**, Pilanci, M., and Pauly, J. Convex Regularization Behind Neural Reconstruction. **ICLR 2021**.

Technical Skills

LLMs and Agents: Agent orchestration, RL environments, coding-agent harnesses, tool calling, SFT, DPO, RLHF
Data and Evaluation: Synthetic-data generation, multi-turn rollouts, evaluation pipelines, confidence calibration
Languages and Frameworks: Python, C/C++, PyTorch, JAX, Hugging Face Transformers
Training and Inference: DeepSpeed, Ray, vLLM
Infrastructure and Experimentation: Linux, Git, Docker, Kubernetes, Microsoft Azure, Google Cloud Platform, Weights & Biases

Selected Honors and Service

- **NSF Graduate Research Fellowship (NSF GRFP)**
- **Stanford EE Departmental Fellowship**
- **Reviewer:** ACL, NeurIPS, ICML, ICLR, MICCAI, IEEE Transactions on Medical Imaging
- **Program Committee:** ICML Workshop on Knowledge and Logical Reasoning in the Era of Data-Driven Learning